

ASRA - Adaptive Scientific Reasoning Architecture

A Fluid-Intelligence Cognitive Architecture for Adaptive Reasoning in Unseen Interactive Environments

Overview

ASRA (Adaptive Scientific Reasoning Architecture) is a modular cognitive architecture designed to investigate fluid intelligence through adaptive reasoning, hypothesis-driven exploration, world-model formation, and reusable abstraction memory.

The project is inspired by the core philosophy behind ARC-AGI-3: true intelligence is not memorization or static prediction, but the ability to rapidly adapt to unfamiliar situations through reasoning, experimentation, and abstraction.

Unlike conventional AI systems that rely heavily on large-scale memorization, fixed policies, or pattern matching, ASRA is designed to behave more like a scientist:

- observing environments,
- forming hypotheses,
- running experiments,
- analyzing outcomes,
- refining internal theories,
- inventing new strategies,
- and transferring reusable reasoning abstractions across tasks.

ASRA explores the idea that fluid intelligence emerges not from static knowledge, but from dynamic cognitive processes operating under uncertainty.

The architecture is intentionally research-oriented and extensible beyond ARC-AGI-3 into broader areas such as:

- cognitive architectures,
 - scientific reasoning systems,
 - adaptive agents,
 - world models,
 - autonomous exploration,
 - embodied intelligence,
 - and future AGI systems research.
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Core Philosophy

ASRA is built around a central belief:

Intelligence is not the memorization of solutions. Intelligence is the adaptive invention of strategies in unfamiliar environments.

Most modern AI systems excel when:

- tasks resemble training distributions,
- objectives are fixed,
- action semantics are known,
- environments are stationary.

ASRA instead focuses on:

- unseen tasks,
- hidden mechanics,
- sparse information,
- uncertain action semantics,
- exploration-driven learning,
- and rapid abstraction formation.

The architecture treats reasoning as an active scientific process rather than passive prediction.

Scientific Reasoning Loop

At the center of ASRA is a recursive scientific reasoning cycle:

Observe

- > Hypothesize
- > Experiment
- > Analyze
- > Refine Theory
- > Retry

This loop governs how the system interacts with novel environments.

Rather than immediately attempting to optimize reward or predict outputs, the system attempts to:

- understand the environment,
 - infer hidden structure,
 - discover causal relationships,
 - and construct internal explanatory models.
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Research Goals

ASRA investigates several core questions related to fluid intelligence:

- Can agents invent new reasoning strategies dynamically?
- Can reusable abstractions emerge from interaction?
- Can systems build compact internal theories of unfamiliar environments?
- Can exploration become hypothesis-driven rather than brute force?
- Can memory store reasoning concepts instead of task-specific solutions?
- Can agents efficiently infer hidden action semantics?
- Can world models improve adaptive problem solving in unseen environments?

The project aims to contribute toward:

- adaptive cognition,
 - scientific reasoning systems,
 - generalization research,
 - and future AGI architectures.
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ARC-AGI-3 Alignment

ASRA is intentionally aligned with the interactive reasoning paradigm introduced in ARC-AGI-3.

ARC-AGI-3 evaluates:

- exploration,
- memory,
- sequential reasoning,
- hidden mechanics,
- adaptive learning,
- and generalization in unseen interactive environments.

ASRA directly targets these capabilities through:

- world-model construction,
- action semantics discovery,
- reflective reasoning,
- hypothesis-driven experimentation,
- and abstraction-centric memory systems.

Rather than treating ARC tasks as static puzzles, ASRA treats them as discoverable environments that must be explored and understood.

High-Level Architecture

ASRA is composed of modular cognitive subsystems that cooperate through an orchestrated reasoning loop.

Core Components

1. Observation Engine

The Observation Engine transforms raw environment frames into structured internal representations.

Responsibilities include:

- object extraction,
- topology analysis,
- connected component detection,
- spatial relationship modeling,
- symmetry detection,
- boundary analysis,
- state transition tracking,
- and environment parsing.

The system converts raw grids into semantically meaningful structures that downstream reasoning modules can operate on.

2. Hypothesis Generation Engine

This module generates candidate explanations for:

- environment behavior,
- action semantics,
- state transitions,
- hidden mechanics,
- and task objectives.

Examples of generated hypotheses:

- "ACTION2 rotates an object."
- "Boundary cells propagate color changes."
- "Object count determines reward state."
- "ACTION6 manipulates local topology."
- "Movement is constrained by object color."

Hypotheses are probabilistic and continuously updated through experimentation.

3. Experimental Simulation Engine

The Simulation Engine performs controlled experiments within the environment.

Responsibilities include:

- testing action consequences,
- evaluating hypotheses,
- generating counterfactual trajectories,
- performing intervention analysis,
- and measuring information gain.

The system actively explores environments to reduce uncertainty and refine its internal models.

This component is heavily inspired by scientific experimentation.

4. Reflective Critic System

The Critic System evaluates:

- hypothesis consistency,
- explanatory power,
- reasoning quality,
- abstraction compactness,
- exploration efficiency,
- and strategy effectiveness.

The critic enables:

- self-evaluation,
- reflection,
- and adaptive refinement.

The system continuously asks:

- "Does this explanation remain consistent?"
 - "Can the reasoning be compressed further?"
 - "Did the experiment reduce uncertainty?"
 - "Should exploration priorities change?"
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5. World Model Layer

ASRA maintains internal predictive models of environment dynamics.

The World Model attempts to learn:

- object behavior,
- state transitions,
- causal relationships,
- interaction rules,
- and action effects.

Rather than directly predicting outputs, the system attempts to understand: how environments evolve under interventions.

This allows:

- planning,
 - simulation,
 - and strategic exploration.
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6. Strategy Invention Module

This is one of ASRA's central research ideas.

Instead of selecting from a fixed set of predefined strategies, the architecture attempts to dynamically invent new reasoning procedures when existing abstractions fail.

Examples:

- recursive decomposition strategies,
- topology-preserving transformations,
- causal graph traversal heuristics,
- hierarchical exploration procedures,
- object-centric planning methods.

This module aims to move beyond: retrieval-based intelligence toward constructive intelligence.

7. Meta-Abstraction Memory System

ASRA includes a reusable abstraction memory system.

Importantly, the memory does NOT store:

- exact task solutions,
- memorized outputs,
- benchmark-specific mappings.

Instead, it stores:

- reasoning strategies,
- exploration patterns,
- abstraction schemas,
- causal motifs,
- failure modes,
- and reusable conceptual structures.

Examples:

- symmetry reconstruction,
- boundary propagation,
- object conservation,
- recursive counting,
- hierarchical decomposition,

- topology repair,
- relational mapping patterns.

This enables transfer learning through conceptual reuse rather than memorization.

8. Planning and Exploration Engine

The Planner coordinates:

- exploration priorities,
- experiment selection,
- hypothesis scheduling,
- uncertainty reduction,
- and action sequencing.

The planner attempts to maximize:

- information gain,
- reasoning efficiency,
- and environment understanding.

Exploration is curiosity-driven rather than random.

Multi-Agent Cognitive Architecture

ASRA can optionally operate as a society of specialized reasoning agents.

Example agent roles include:

Agent	Responsibility
Observer Agent	Parse environment structure
Scientist Agent	Generate hypotheses
Experiment Agent	Test interventions
Critic Agent	Evaluate explanations
Memory Agent	Retrieve abstractions
Planner Agent	Coordinate exploration
Reflection Agent	Analyze failures
Compression Agent	Search for compact explanations

This modular structure enables:

- parallel reasoning,
 - specialization,
 - iterative refinement,
 - and scalable cognitive orchestration.
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Action Semantics Discovery

A key challenge in ARC-AGI-3 is that action meanings vary between environments.

ASRA treats action understanding as a scientific discovery problem.

The architecture attempts to infer:

- what each action does,
- when actions are valid,
- causal consequences,
- spatial effects,
- environment constraints,
- and interaction mechanics.

The system builds evolving internal theories of action semantics through experimentation.

Intrinsic Curiosity and Information Gain

ASRA prioritizes experiments that maximize:

- uncertainty reduction,
- causal discovery,
- state-space understanding,
- and environment compression.

The system continuously asks:

- "Which action teaches me the most?"
- "Which hypothesis most reduces uncertainty?"
- "Which experiment best separates competing theories?"

This produces more intelligent exploration behavior than brute-force search.

Long-Term Vision

ASRA is not intended to be merely a competition submission.

The broader vision is to explore:

- fluid intelligence,
- adaptive cognition,
- reusable abstraction learning,
- scientific reasoning architectures,
- and world-model-based intelligence.

The project may evolve into:

- research papers,
 - open-source cognitive architecture frameworks,
 - adaptive agent platforms,
 - scientific reasoning systems,
 - or broader AGI research infrastructure.
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Technical Stack (Initial Direction)

Core Runtime

- Python

Numerical / ML

- PyTorch
- NumPy
- JAX (optional later)

Graph / Symbolic Systems

- NetworkX
- Z3
- symbolic reasoning libraries

Agent Orchestration

- custom planner/scheduler
- multi-agent coordination framework

Environment Interaction

- ARC-AGI-3 toolkit
- custom experimentation engine

Visualization

- reasoning traces
 - hypothesis trees
 - exploration maps
 - world-model inspection tools
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Guiding Principles

ASRA is guided by several principles:

Adaptation over memorization

The system must invent solutions dynamically.

Scientific reasoning over brute-force search

Exploration should be hypothesis-driven.

Abstraction over task-specific optimization

Memory stores concepts, not answers.

Reflection over static policies

The system continuously critiques itself.

World understanding over output prediction

The architecture builds internal explanatory models.

Fluid intelligence over benchmark fitting

Generalization is the primary objective.

Project Positioning

ASRA can be described as:

A fluid-intelligence cognitive architecture for adaptive scientific reasoning, exploration, and strategy invention in unseen interactive environments.

Or more concisely:

ASRA investigates whether intelligence can emerge from scientific reasoning loops operating over reusable abstractions, world models, and adaptive exploration.

GitHub Repo Name: asra

Full Framework Name: ASRA - Adaptive Scientific Reasoning Architecture

Tagline: A fluid-intelligence architecture for adaptive scientific reasoning in unseen interactive environments.